

Houssam HADHADI

☎ +212 6 39 90 77 11 — ✉ houssamhadhadi@gmail.com — [in](#) LinkedIn — [GitHub](#)

Profil

Full-stack Software Engineer with a solid frontend foundations in React, and JavaScript, complemented by backend development with Node.js and Express. Experienced in deploying containerized infrastructure and developing production-ready applications with REST APIs, Docker, Linux, and networking. Graduate of 1337 Coding School (42 Network), a project-based engineering program focused on problem-solving and peer-reviewed learning.

An autonomous and adaptable engineer with experience in collaborative development, peer code reviews, and project delivery. Seeking a Software Engineer Internship to contribute to an engineering-driven team.

Experience

Network Security Intern

Alcanet

Marrakech, Morocco

Jun 2023 – jul 2023

- Configured PfSense as a network firewall, established VLANs for traffic segmentation, and administered NAT, DHCP, and DNS services.
- Designed and implemented a DMZ to isolate public-facing services, and deployed Squid with SquidGuard for proxy-based URL filtering using blacklists and ACLs.
- Monitored system logs and documented network configurations for internal reference.

Systems & Network Intern

Digital Union

Marrakech, Morocco

Jun 2021 – jul 2021

- Implemented a Hyper-V Failover Cluster with a multi-node topology, shared storage, and Active Directory integration.
- Managed disks, volumes, and network interfaces using Windows Failover Cluster Manager, and validated service continuity through simulated node failures.
- Configured network infrastructure using UniFi, performed structured cabling, and installed IP surveillance cameras.

Technical Projects

Transcendence – Full Stack Travel Platform *Next.js, TypeScript, Node.js, REST APIs,*

Docker

- Built responsive, dynamic UIs with React, Next.js, and TypeScript in a component-based architecture, delivering 20+ reusable components across the application.
- Implemented advanced features including multi-criteria search, trip planning workflows, and a real-time social feed.
- Integrated REST APIs for efficient client-server communication and reliable asynchronous data handling.
- Developed and deployed the full stack using Docker Compose in an isolated Linux environment, applying a DevOps approach to manage service orchestration and networking.
- Managed full Git workflow with feature branches, pull requests, and peer code reviews; participated in regular team meetings to discuss tasks and align on technical decisions.

Movie Discovery Web Application *React, TypeScript, TailwindCSS, REST APIs*

- Built a React and TypeScript movie discovery app consuming the TMDB API, leveraging JavaScript (ES6+) with dynamic search and multi-criteria filtering, optimized via lazy loading.
- Architected scalable state management using React Context API and client-side routing for a fast, maintainable single-page application.
- Designed a mobile-first, accessible UI using HTML5, CSS3, and TailwindCSS with smooth interactions and cross-browser compatibility.

HTTP Web Server *C++, HTTP, Nginx-inspired Architecture, epoll*

- Developed a production-grade HTTP server in C++ implementing full request parsing, routing, and HTTP/1.1 protocol handling.
- Implemented event-driven I/O using epoll, enabling concurrent handling of 100+ simultaneous client connections without performance degradation.

- Designed a configurable architecture supporting virtual hosts, custom routes, and error management, demonstrating strong analytical skills by simulating Nginx behavior to create robust configuration files.

Containerized Web Infrastructure *Docker, Docker Compose, Nginx, MariaDB, Redis*

- Designed and deployed a complete multi-container infrastructure (Nginx, WordPress, MariaDB, Redis) with isolated networking and persistent volumes.
- Applied DevOps fundamentals, ensuring security by configuring SSL/TLS and hardening the infrastructure while managing containerization and service orchestration.
- Proactively managed incident resolution by monitoring container health and service logs to ensure high availability of the infrastructure.

Minishell *C, Unix, Process Management, Pipes*

- Built a Unix shell in C supporting command execution, pipes, redirections, and environment variable handling.
- Implemented process management using fork, exec, wait, and signal handling, deepening knowledge of OS internals and low-level programming.

Technical Skills

Frontend: React.js, Next.js, JavaScript (ES6+), TypeScript, HTML5, CSS3, TailwindCSS, responsive design, mobile-first, accessibility (a11y), DOM manipulation, cross-browser compatibility, Jest, Testing Library.

Backend: Node.js, Express.js, REST APIs.

Web & Networking: HTTP protocol, client-server architecture, request/response lifecycle, web server internals, network administration.

DevOps & Infrastructure: Docker, Docker Compose, Nginx, Linux CLI, CI/CD (fundamentals), SSL/TLS configuration, containerized environments.

Tools: Git, GitHub, Postman, Linux CLI.

Design & UI: Figma, Balsamiq, Fundamentals of UI design.

Education

1337 Coding School (42 Network)

Software Engineering — Low-level Programming, Web Development, Systems, Algorithms

Benguerir, Morocco

2023 – 2026

Faculty of Sciences Semlalia

Bachelor's Degree (R2SI) — Networks, Security, and Computer Systems

Marrakech, Morocco

2022 – 2023

OFPPT

Specialized Technician — Computer Systems and Network Administration

Marrakech, Morocco

2019 – 2021

Lycee Zerktouni

High School Diploma (Baccalaureate) — Physical Sciences

Marrakech, Morocco

2018 – 2019